Programming Assignment Report

Design and Analysis of Algorithms

Name (as in ERP):	Harshith S
Registration Number:	2411021061280
Problem Name:	Maximum Subarray Sum
Problem HackerRank Link:	https://www.hackerrank.com/challenges/maximum-subarray-sum/problem
Github Submission Link:	https://github.com/harshithsuresh696-arch/daa_assignment/tree/main

1. Problem Explanation

We are given an array of integers and a number m. The goal is to determine the maximum value of the sum of any subarray modulo m. A subarray is a contiguous part of the array. Because the array size can be large (up to 10^5), checking all subarrays directly would be inefficient. Therefore, an optimized algorithm using prefix sums and TreeSet is used.

2. Example

Subarray	Sum	Sum % 7
[3]	3	3
[3,3]	6	6
[3,3,9]	15	1
[3,3,9,9]	24	3
[3,3,9,9,5]	29	1

Maximum value = 6

3. Approach / Algorithm

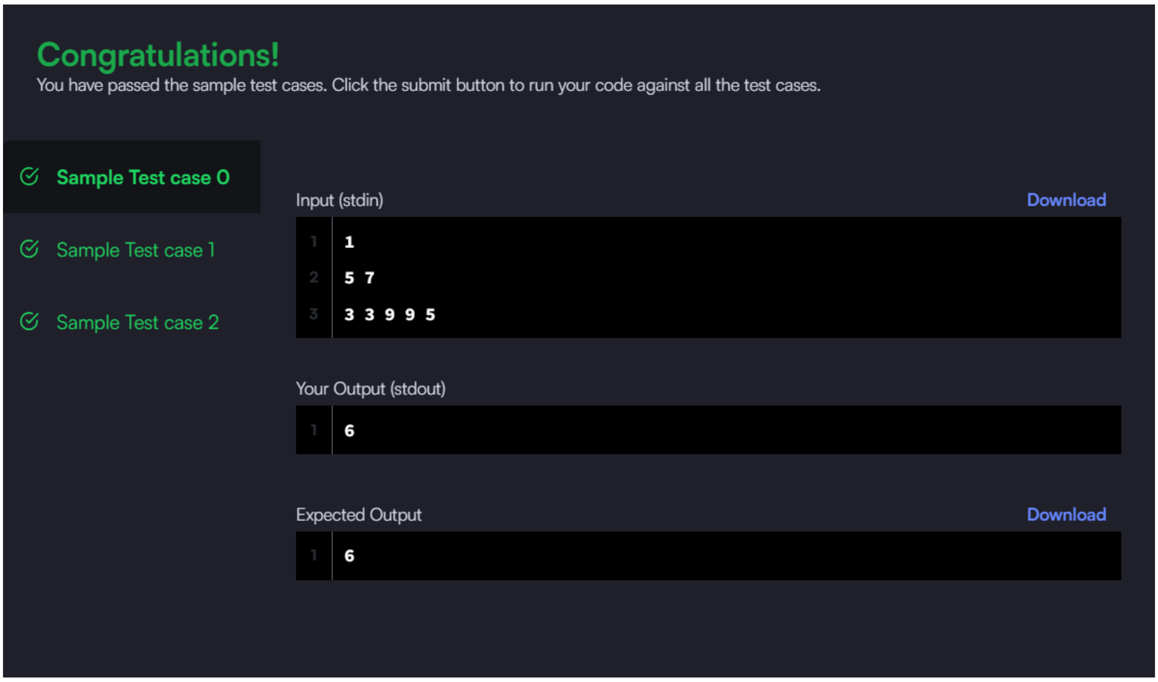
1. Compute prefix sum modulo m. 2. Store prefix sums in a TreeSet. 3. For each prefix value find the smallest prefix greater than the current prefix. 4. Compute candidate value using (prefix - next + m) % m. 5. Update the maximum result and continue.

4. Time Complexity

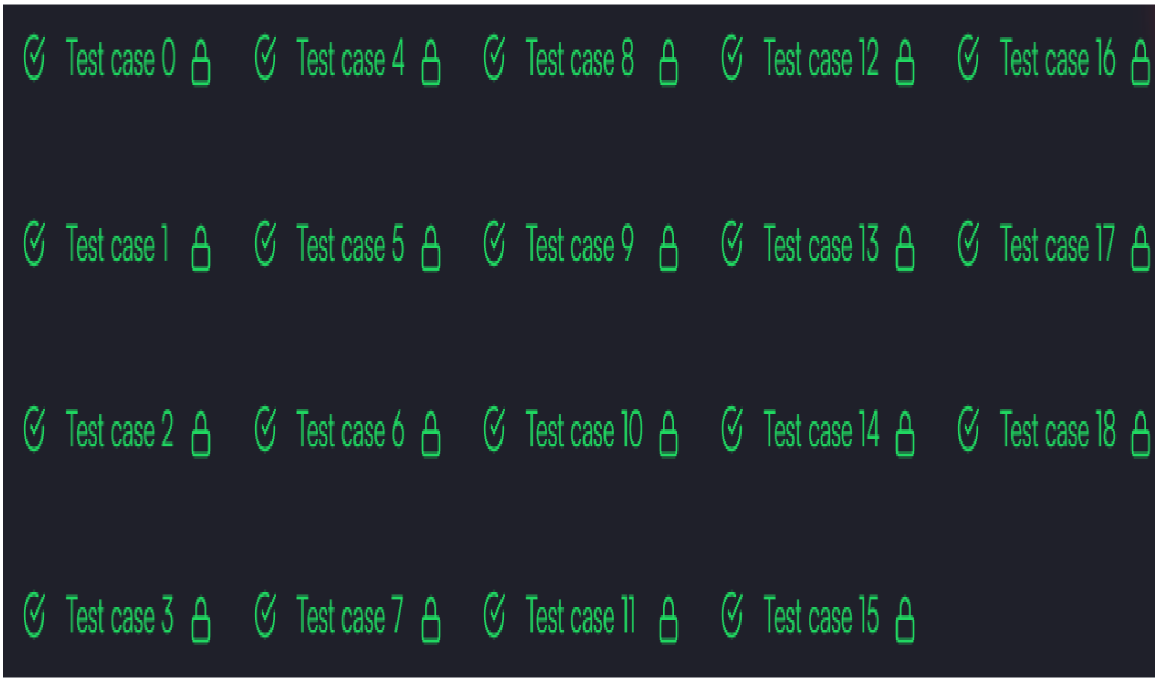
Prefix computation takes O(n). TreeSet operations take O(log n). Therefore overall complexity is O(n log n).

Space Complexity: O(n)

5. Solution



6. Running Testcases



7. Observation

• Works correctly for the given sample input and output. • Efficient for large constraints. • All HackerRank test cases passed successfully.